1. How many pins does the 8051 microcontroller have?

A) 20
B) 30
C) 40
D) 50

✅ **Answer:** C) 40

🔹 **2. What is the function of Pin 9 (RST) in the 8051 microcontroller?**

A) Enables serial communication
B) Interrupt pin
C) Resets the microcontroller
D) Power supply

✅ **Answer:** C) Resets the microcontroller

🔹 **3. Which pin is used to connect the ground in 8051?**

A) Pin 40
B) Pin 30
C) Pin 20
D) Pin 10

✅ **Answer:** C) Pin 20

🔹 **4. What is the use of the ALE pin (Pin 30) in 8051?**

A) Enables interrupts
B) Latches address from data
C) Resets the microcontroller
D) Enables program memory

✅ **Answer:** B) Latches address from data

🔹 **5. Which port is used for multiplexed address/data lines?**

A) Port 0
B) Port 1
C) Port 2
D) Port 3

✅ **Answer:** A) Port 0

🔹 **6. What does PSEN stand for?**

A) Power Signal Enable
B) Program Signal Enable
C) Program Store Enable
D) Peripheral Store Enable

✅ **Answer:** C) Program Store Enable

🔹 **7. What is the function of EA (Pin 31)?**

A) Reset microcontroller
B) Enable external memory access
C) Start serial communication
D) Provide clock signal

✅ **Answer:** B) Enable external memory access

🔹 **8. Which port in 8051 has only general-purpose I/O and no alternate function?**

A) Port 0
B) Port 1
C) Port 2
D) Port 3

✅ **Answer:** B) Port 1

🔹 **9. What is the function of XTAL1 and XTAL2?**

A) Provide power
B) Provide ground
C) Clock oscillator inputs
D) Interrupt control

✅ **Answer:** C) Clock oscillator inputs

🔹 **10. Port 3 pin P3.0 is used as:**

A) WR
B) RD
C) TXD
D) RXD

✅ **Answer:** D) RXD

🔹 **11. Which port also serves as the high-order address bus during external memory access?**

A) Port 0
B) Port 1
C) Port 2
D) Port 3

✅ **Answer:** C) Port 2

🔹 **12. What is the voltage level typically required at the Vcc pin (Pin 40)?**

A) 3.3V
B) 6V
C) 5V
D) 9V

✅ **Answer:** C) 5V

🔹 **13. Which port is open-drain and requires external pull-up resistors for I/O operation?**

A) Port 0
B) Port 1
C) Port 2
D) Port 3

✅ **Answer:** A) Port 0

🔹 **14. P3.2 and P3.3 are used for which alternate functions?**

A) TXD and RXD
B) INT0 and INT1
C) WR and RD
D) T0 and T1

✅ **Answer:** B) INT0 and INT1

🔹 **15. What is the role of P3.6 and P3.7 in Port 3?**

A) Timer inputs
B) Interrupt inputs
C) Serial communication
D) External memory write/read control

✅ **Answer:** D) External memory write/read control

🔹 **16. In a typical 8051, which pin controls whether internal or external program memory is used?**

A) PSEN
B) EA
C) ALE
D) RST

✅ **Answer:** B) EA

🔹 **17. Which pin is active low in 8051?**

A) VCC
B) PSEN
C) ALE
D) EA

✅ **Answer:** B) PSEN

🔹 **18. Which pins are used for connecting a crystal oscillator?**

A) P1.0 and P1.1
B) XTAL1 and XTAL2
C) EA and ALE
D) RD and WR

✅ **Answer:** B) XTAL1 and XTAL2

🔹 **19. Port 3.4 and Port 3.5 are used as:**

A) TXD and RXD
B) INT0 and INT1
C) T0 and T1
D) WR and RD

✅ **Answer:** C) T0 and T1

🔹 **20. What is the function of P3.1 (Pin 11)?**

A) TXD
B) RXD
C) T0
D) INT0

✅ **Answer:** A) TXD

Would you like these as a **printable PDF**, or a **quiz format** with scoring and randomization for practice?

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Here are **20 more multiple-choice questions (MCQs)** on the **8051 microcontroller pin diagram and related concepts**, including internal functionality, memory interface, and port usage.

🧠 **Advanced MCQs on 8051 Microcontroller (Set 2)**

🔹 **21. The alternate function of P3.5 is:**

A) T0
B) T1
C) RXD
D) WR

✅ **Answer:** B) T1

🔹 **22. The 8051 uses how many pins for I/O operations in total?**

A) 16
B) 24
C) 32
D) 40

✅ **Answer:** C) 32
(Ports 0, 1, 2, and 3 — each 8 bits)

🔹 **23. What does the ALE signal do when interfacing with external memory?**

A) Enables internal ROM
B) Selects external RAM
C) Latches lower address byte
D) Acts as clock generator

✅ **Answer:** C) Latches lower address byte

🔹 **24. Which control signal is required to read from external program memory?**

A) ALE
B) PSEN
C) WR
D) RD

✅ **Answer:** B) PSEN

🔹 **25. What type of signal is required at RST to reset the 8051?**

A) Active low pulse
B) Constant high signal
C) Active high pulse (>2 machine cycles)
D) Active edge-triggered pulse

✅ **Answer:** C) Active high pulse (>2 machine cycles)

🔹 **26. Which port is bidirectional and does not have alternate functions?**

A) Port 0
B) Port 1
C) Port 2
D) Port 3

✅ **Answer:** B) Port 1

🔹 **27. What is the role of P3.6 and P3.7 in memory operations?**

A) Timer inputs
B) Clock outputs
C) Write and Read control signals
D) Reset and Enable

✅ **Answer:** C) Write and Read control signals

🔹 **28. Which of the following ports requires pull-up resistors when used as I/O?**

A) Port 1
B) Port 2
C) Port 0
D) Port 3

✅ **Answer:** C) Port 0

🔹 **29. PSEN is activated when:**

A) Reading from internal RAM
B) Reading from external ROM
C) Writing to I/O ports
D) Resetting the microcontroller

✅ **Answer:** B) Reading from external ROM

🔹 **30. What happens when EA is connected to GND?**

A) Internal memory is used
B) External program memory is used
C) ALE is disabled
D) All ports become inputs

✅ **Answer:** B) External program memory is used

🔹 **31. Which pins of Port 0 carry data when accessing external memory?**

A) P0.0 to P0.3
B) P0.4 to P0.7
C) P0.0 to P0.7
D) P1.0 to P1.7

✅ **Answer:** C) P0.0 to P0.7

🔹 **32. Which pins are used for serial communication?**

A) P3.2 and P3.3
B) P3.0 and P3.1
C) P0.0 and P0.1
D) XTAL1 and XTAL2

✅ **Answer:** B) P3.0 and P3.1 *(RXD and TXD)*

🔹 **33. Which interrupt pins are present in Port 3?**

A) INT0 and INT1
B) INT2 and INT3
C) T0 and T1
D) WR and RD

✅ **Answer:** A) INT0 and INT1 *(P3.2 and P3.3)*

🔹 **34. Which pin is used to latch the address during external memory access?**

A) PSEN
B) ALE
C) EA
D) XTAL1

✅ **Answer:** B) ALE

🔹 **35. In the 8051, the clock frequency typically used is:**

A) 1 MHz
B) 5 MHz
C) 12 MHz
D) 16 MHz

✅ **Answer:** C) 12 MHz

🔹 **36. Which port is also used to send the upper 8 bits of the 16-bit address during external memory access?**

A) Port 0
B) Port 1
C) Port 2
D) Port 3

✅ **Answer:** C) Port 2

🔹 **37. Which pin enables external devices to fetch code from external ROM?**

A) ALE
B) EA
C) PSEN
D) WR

✅ **Answer:** C) PSEN

🔹 **38. What is the alternate function of P3.4?**

A) RXD
B) INT0
C) T0
D) WR

✅ **Answer:** C) T0

🔹 **39. If EA = 1, the microcontroller will:**

A) Use only external ROM
B) Use internal ROM only
C) Use internal ROM unless external is specified
D) Disable all program memory access

✅ **Answer:** C) Use internal ROM unless external is specified

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🔹 **40. Which of the following ports do NOT support alternate functions?**

A) Port 0
B) Port 1
C) Port 2
D) Port 3

✅ **Answer:** B) Port 1